

 AI-POWERED CYCLONE RESILIENCE PLATFORM

PRAHAR

Predictive Risk & Anticipatory Hazard Action Resource

Track-Based Cyclone Impact & Infrastructure Vulnerability Forecaster for Bay of Bengal & Coastal APAC

Google Earth Engine

Gemini 1.5 Flash

FastAPI & React

22 Indic Languages

Designed for District Magistrates, NDRF Commanders & State Disaster Authorities (OSDMA, WBDMD, APSDMA, TNDMA) to trigger anticipatory evacuations < T-12h before landfall.

Core Objectives

1. Track Telemetry

Ingest IMD, JTWC, Tide Gauges & Copernicus 30m GEE DEM in real-time.

2. Surge Modeling

Hydrodynamic coastal inundation envelopes across 3 surge hazard tiers.

3. Infrastructure Graph

Pinpoint severed CHCs, tripped substations, and cutoff shelters.

4. Multimodal SOPs

Gemini Flash auto-drafts tactical cadre directives in 22 Indic tongues.

5. Parametric Escrow

Automated smart escrow releases disaster liquidity at T-24h with 0 latency.

Our Mission: Transform reactive cyclone disaster relief into predictive, automated anticipatory action — protecting 170M+ coastal citizens across the Bay of Bengal & Coastal APAC.

The Coastal Vulnerability Gap

The Problem Today

▲ Fragmented Telemetry Silos

Cyclone tracks from IMD, bathymetry, power grids, and hospital directories exist in disconnected silos. No real-time spatial correlation.

▲ 12-24h Bureaucratic Response Lag

Current administrative dashboards operate post-landfall. Damage assessments occur 24-48 hours after coastal embankments are overtopped.

▲ Static Paper-Based Emergency SOPs

District Collectors rely on generic paper manuals unsuited for live dynamic water levels, leading to trapped civilians and submerged relief routes.



"The storm surge inundated the coastal arterial highway before the evacuation buses were dispatched."

Without predictive, track-based infrastructure vulnerability modeling, relief forces deploy blindly after lifelines sever. Coastal districts lose critical power and emergency medical access within hours.

The PRAHAR Solution

1. Unified Geoprobe

Direct ingestion of live IMD track coordinates, JTWC advisories, Copernicus 30m DEM elevation, and OSM critical lifelines into a unified spatial graph.

2. Multimodal AI Triage

Gemini 1.5 Flash evaluates hydrological choke points, drafting role-directed SOPs and generating multi-lingual Web Speech broadcasts in 22 Indic languages.

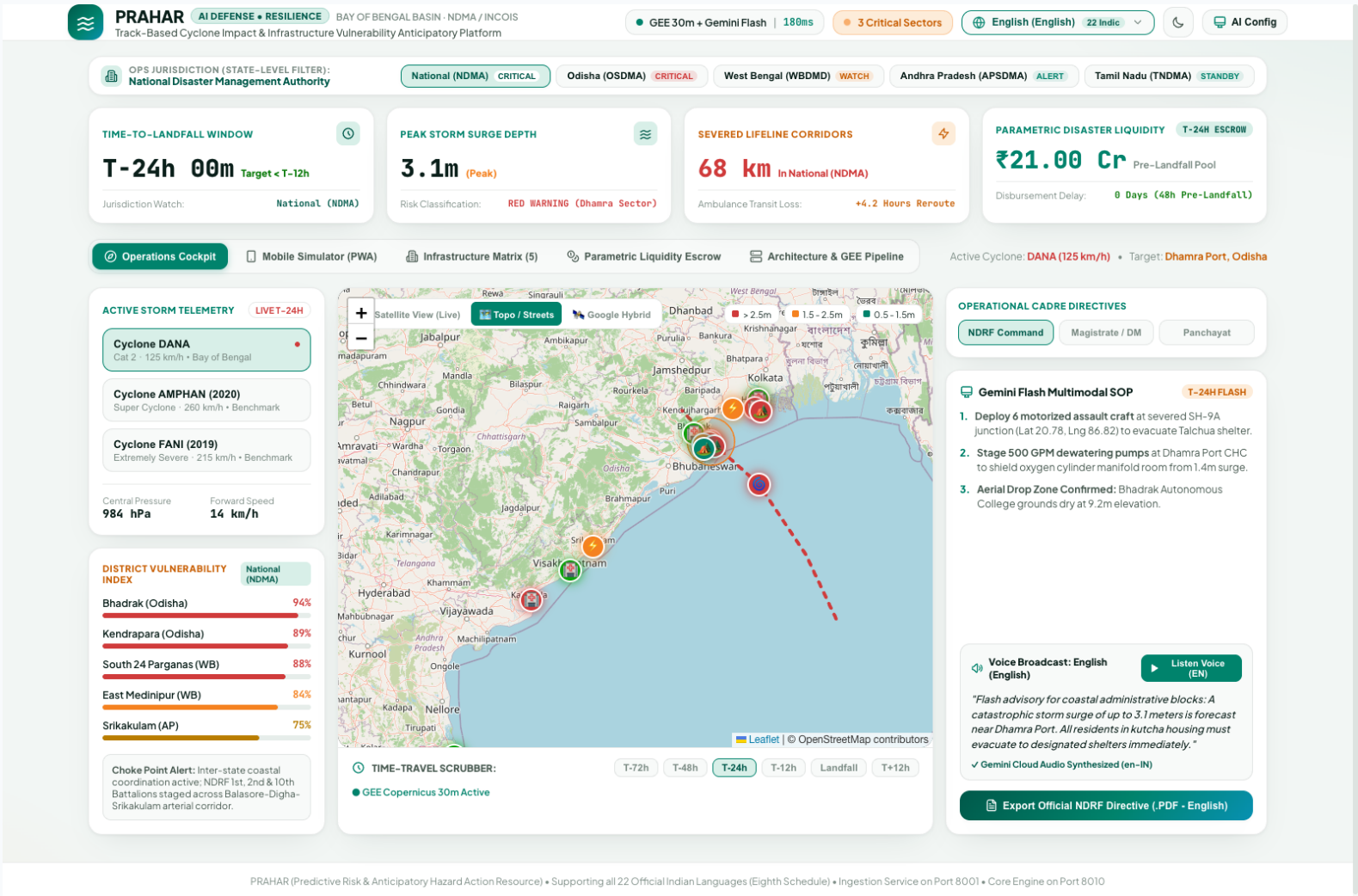
3. Parametric Liquidity

Automated pre-landfall escrow releases emergency cash reserves at T-24h directly to SDRF accounts without bureaucratic verification delays.



The 5-Minute District Collector Value: You don't have to decipher raw radar files. PRAHAR instantly highlights which bridge is severed, which hospital loses backup power, and where assault craft must be pre-staged.

Live Operations Cockpit



Operational Capabilities

• Real-time Track Ingestion

Tracks active storms (e.g. Cyclone DANA 125 km/h) with 6-stage temporal waypoints.

• 3-Tier Surge Inundation

High-resolution polygons for >2.5m, 1.5-2.5m, and 0.5-1.5m forecast surge.

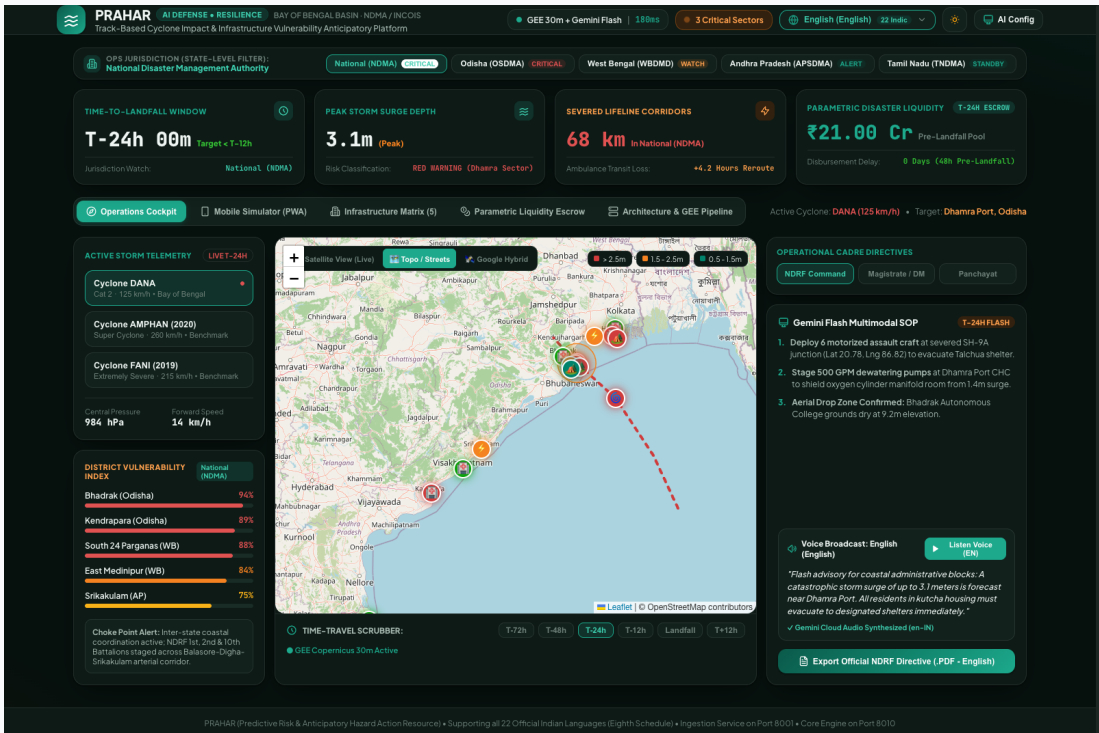
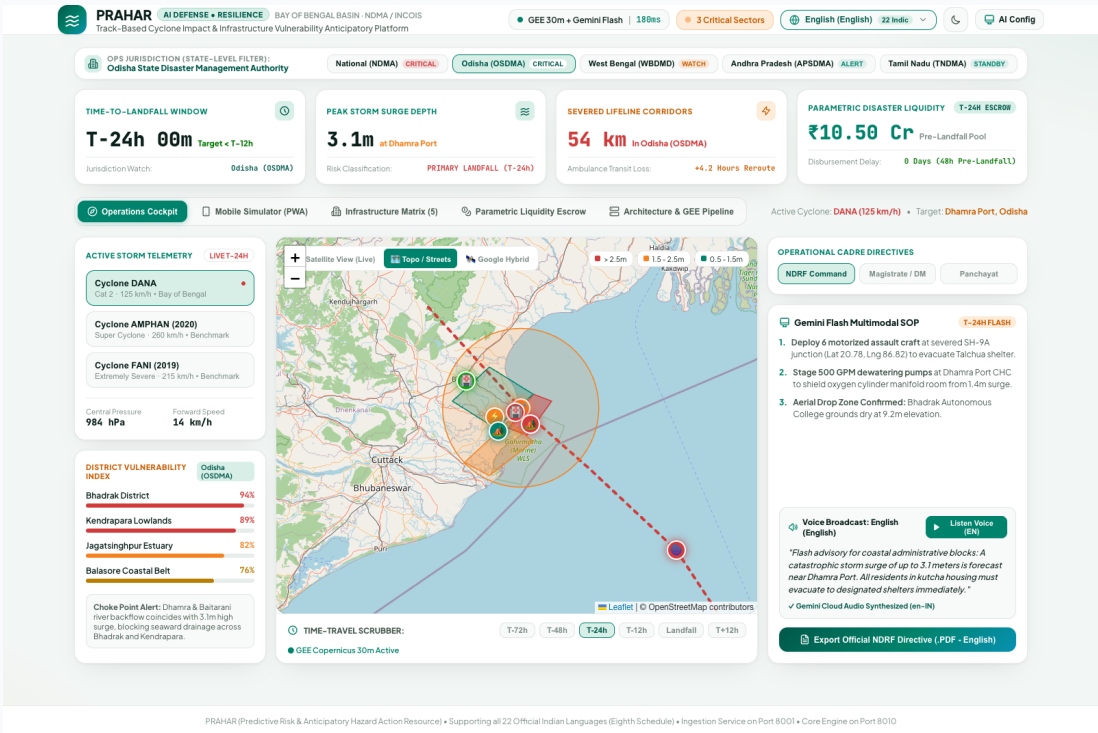
• District Vulnerability Index

Automated ranking of districts (Bhadrak 94%, Kendrapara 89%).

• Multimodal Directives

Gemini 1.5 Flash generates action orders for NDRF, DMs & Panchayats.

State Operations & 22 Official Indic Languages



Multi-State Operational Jurisdictions

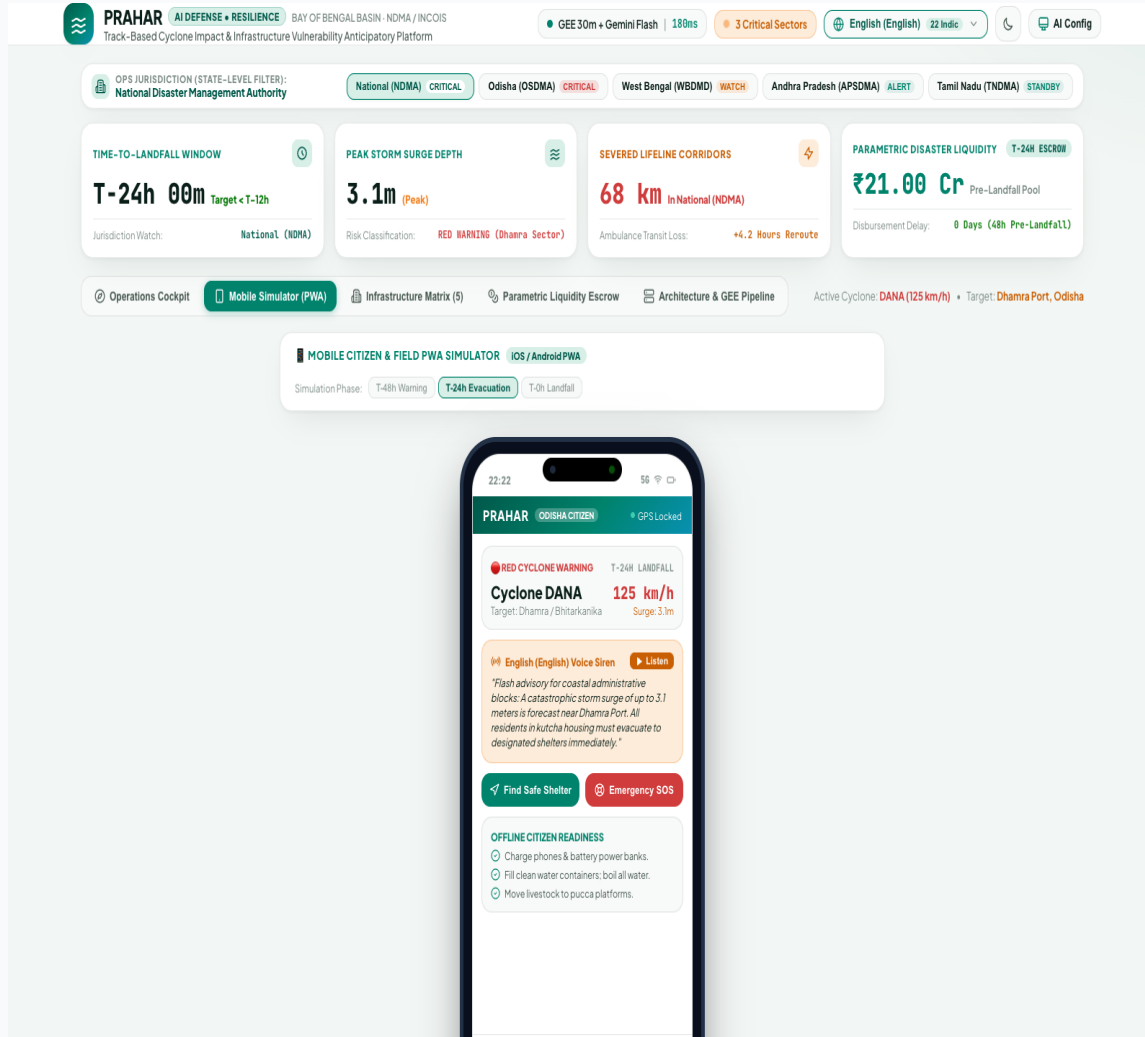
1-click filtering across National (NDMA), Odisha (OSDMA), West Bengal (WBDMD), Andhra Pradesh (APSDMA), and Tamil Nadu (TNDMA) automatically re-centers GIS and updates state-specific choke points.



All 22 Eighth Schedule Indian Languages

Native translations and instant Web Speech audio broadcasts for Odia, Bengali, Telugu, Tamil, Hindi, Marathi, Gujarati, etc. Dark Mode supports 24/7 command center screening without visual fatigue.

Mobile Citizen Simulator (PWA)



Last-Mile Citizen Protection

✓ Offline-First Progressive Web App

Pre-caches elevation maps and shelter waypoints. Fully functional even when mobile towers lose power or backhaul connections.

✓ Safe Navigation to Multi-Purpose Shelters

Real-time routing diverts citizens away from flooded roads toward elevated shelters with verified remaining capacity.

✓ One-Tap SOS Distress Beacon

Broadcasts GPS coordinates, battery level, and family count to district emergency operations centers.

✓ Parametric Relief QR Pass

Digital QR credentials enable immediate ration and relief fund verification at local evacuation shelters.

100



Primary Health Centers (CHCs)

33/11kV Substation Grids

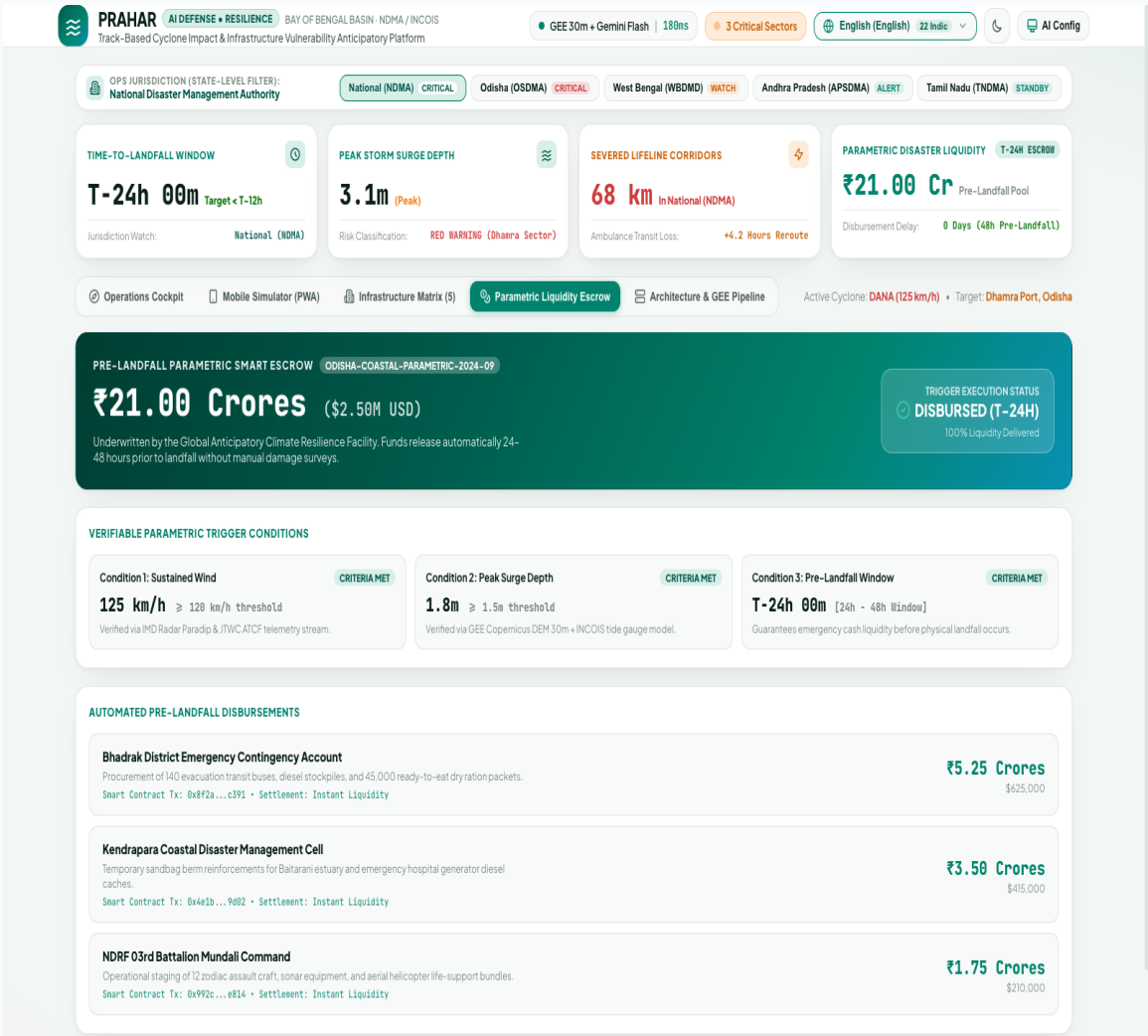
Multi-Purpose Cyclone Shelters



Arterial Highways & Bridges

- Identifies specific road cuts (e.g. SH-9A) causing ambulance reroutes of 4.2+ hours.

Parametric Escrow & Predictive Mathematical Model



$$V_dist = 0.45 * (S_peak / E_mean) + 0.35 * (C_sev / C_tot) + 0.20 * \rho_kutchra$$

Mathematical Vulnerability Formulation

Escrow Automation Protocol:

- Zero Claims Processing: Funds disburse automatically at T-24h once storm track & surge thresholds are met.
- Pre-Landfall Disaster Liquidity: Allocates ₹21.00 Cr pool directly to impacted coastal state SDRF accounts (e.g. ₹10.5 Cr Odisha, ₹5.25 Cr WB).
- Cryptographic Auditability: Every payout is recorded with verifiable blockchain-inspired hashes preventing administrative diversion.

System Architecture & Decoupled Pipelines

1. External Feeds & Telemetry

- IMD / JTWC: Real-time cyclone tracks & cones
- Google Earth Engine: Copernicus DEM 30m, Sentinel-1 SAR
- OSM Overpass: Hospitals, Sub-stations, Arterial Bridges
- INCOIS / NOAA: Coastal Tide Gauges & Bathymetry

2. Ingestion Microservice (:8001)

- Async Ingestion Worker & Poller Daemon
- Track Normalizer & Cone of Uncertainty Generator
- GEE Elevation & Bathymetric Sampler
- OSM Infrastructure Feature Extractor

3. Core Simulation Engine (:8010)

- FastAPI 0.111 Application Core
- SLOSH Storm Surge & Inundation Model
- Compound River-Surge Backwater Engine
- NetworkX Infrastructure Severance Graph
- Parametric Escrow Controller & Gemini AI

4. Storage & Presentation Tier

- PostgreSQL / PostGIS Spatial DB & Redis Cache
- React 18 + TS + Leaflet GIS 3D Cockpit (:5174)
- T-72h to Landfall Temporal Playback Scrubber
- Role-Differentiated SOPs & 22 Indic Audio

⚡ Decoupled Pipeline: External Feeds → Ingestion Worker (:8001) → Spatial DB → Core Engine (:8010) → React GIS (:5174)

End-to-End System Block Diagram

1. EXTERNAL FEEDS & SATELLITE TELEMETRY



IMD / JTWC

Cyclone track coordinates, wind & pressure



Google Earth Engine

Copernicus 30m DEM, Sentinel-1 & GPM



OSM Overpass

Hospitals, Sub-stations, Bridges & Shelters



INCOIS / NOAA

Coastal tide gauges, SST & bathymetry

2. DATA INGESTION MICROSERVICE (services/ingestion :8001)



Async Worker : Non-blocking poller daemon



Track Normalizer : Cone generator



Raster Pipeline : GEE DEM sampler



OSM Extractor : Spatial lifelines parser

3. STORAGE & GEO-CACHING TIER



PostgreSQL / PostGIS — Spatial DB & Vector Geometries



Redis Geo-Cache & Pub/Sub — Real-time Telemetry Stream Cache



GeoTIFF & GeoJSON Store — Inundation Polygons & Elevation Grids

4. CORE API & SIMULATION ENGINE (backend FastAPI :8010)



SLOSH Surge Model

Hydrodynamic surge & bathymetry



Compound Flood

River backflow & compound runoff



NetworkX Graph

Dynamic road severance & isolation



Parametric Escrow

T-24h smart contract escrow trigger



Gemini Flash AI

Multimodal reasoning & 22 Indic SOPs

5. PRESENTATION LAYER (frontend React 18 + TypeScript + Vite :5174)



War Room Console

National & State ops command



Leaflet GIS 3D

Inundation & severed lifelines



Temporal Scrubber

T-72h to Landfall playhead



Indic Audio Drawer

22 Official languages speech



Parametric Terminal

Pre-landfall liquidity release

Real-Time Pipeline: Feeds & GEE → Ingestion (:8001) → PostGIS / Redis → FastAPI Engine (:8010) → SLOSH & Gemini Flash → War Room (:5174)